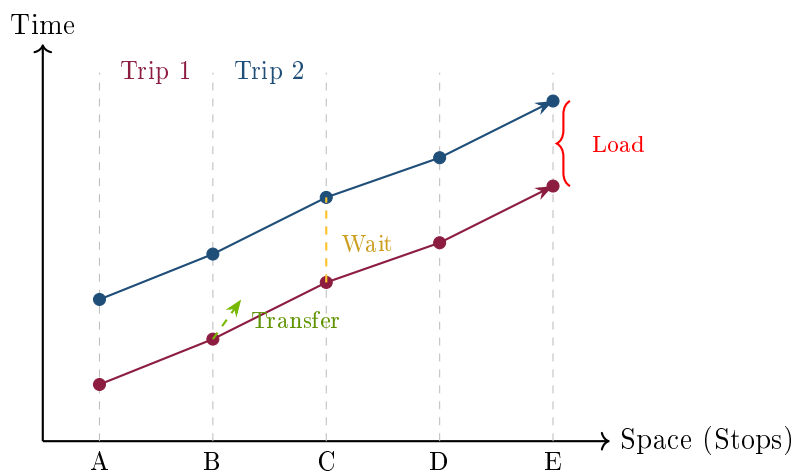


GTFS2GMNS

Transit System Modeling

From Data Conversion to Decision Variables



Course Specification & Deliverable Requirements

Spring 2026

Core Principle: GTFS files are not static data—
they are **decision variables**.

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Version 1.0 — January 25, 2026

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1 Introduction: Beyond Node-Link Visualization

The fundamental principle of this course is transformational: **GTFS files are not static data—they are decision variables.**

Traditional approaches treat `stops.txt` as dots on a map and `stop_times.txt` as a fixed timetable. This course requires students to understand these files as the *output* of planning decisions that can be *optimized*.

1.1 Required Reading Materials

Essential Course Documents — READ BEFORE STARTING

Before beginning any assignments, students **must** read and understand the following three documents:

1. Lecture 2.1: Valley Metro Service Planning (Joe Gregory, Valley Metro)

- Real-world service planning and GIS operations
- Origin-Destination study methodology (17,000 surveys)
- **Case Study: Fiesta BUZZ** — interlining and blocking
- Autonomous vehicle pilot integration
- *Key slides: 13–16 (Scheduling/Blocking), 8–12 (OD Study)*

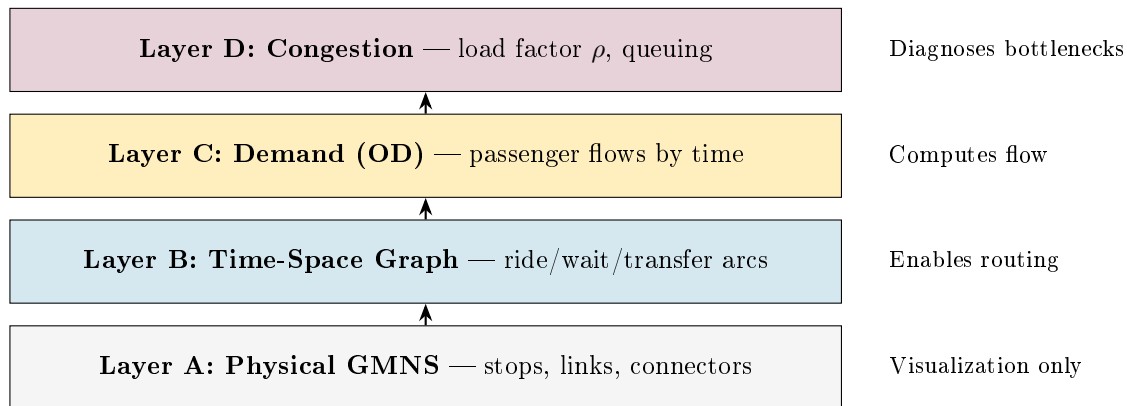
2. Lecture 2.2: Bus and Driver Scheduling (Short Overview)

- Vehicle blocking and crew scheduling fundamentals
- Cycle time and fleet size calculations
- Deadhead minimization principles
- Layover and recovery time requirements

3. Lecture 2.4: GTFS2GMNS User Guide (ASU Trans+AI Lab)

- Complete technical reference for data conversion
- Node types: Physical nodes vs. Service nodes
- Link types: Service, Boarding, Deboarding, Transfer
- Applications: Traffic assignment, LP formulations, accessibility
- *Key sections: 4 (Illustrative Example), 6 (Applications)*

1.2 The Four-Layer Framework



Critical Understanding

Without Layer B (Time-Space Graph), students cannot answer *any* operational question. Without Layer D (Congestion), they cannot answer “where is it crowded and why?” Simply producing `node.csv` and `link.csv` is **not** transit system modeling.

2 Phase 1: Digital Anatomy (Measurable Data Mastery)

Students must prove they understand *how* GTFS files create the time-space representation. This phase establishes the “why” behind every data element.

Required Reading for Phase 1

Primary: Lecture 2.4 — GTFS2GMNS User Guide, Sections 1–4

Reference: GTFS Official Specification (gtfs.org)

2.1 Module 1.1: The Stop-Link-Route Chain

Deliverable 1.1: Physical Network Generation

Input Files: `stops.txt`, `shapes.txt`, `routes.txt`

Task: Use GTFS2GMNS to generate the physical network

Output: Clean GMNS CSVs (`node.csv`, `link.csv`)

Verification Checklist:

- ☐ Number of physical nodes matches unique stops in `stops.txt`
- ☐ All routes are represented in service links
- ☐ Node coordinates are valid (lat/lon within expected range)
- ☐ Link geometries are properly formatted (WKT LineString)

Metric 1.1: Circuitry Index

The **Circuitry Index** measures route directness:

$$\text{Circuitry} = \frac{\text{Route Length (along shape)}}{\text{Euclidean Distance (origin to terminus)}}$$

Calculation Steps:

1. Extract route shape from `shapes.txt`
2. Calculate total route length using Haversine formula between consecutive points
3. Calculate straight-line distance between first and last stop
4. Compute ratio

Interpretation:

Value	Interpretation
≈ 1.0	Direct express service
1.2–1.5	Typical urban route
> 2.0	Highly circuitous (coverage-oriented)

Question to Answer: Why does the bus deviate? Is it for coverage (serving more people) or is there a constraint (road network, terrain)?

2.2 Module 1.2: Timetable to Service-Node Expansion

This module connects to the **two-layer network structure** described in Lecture 2.4 (Section 4.1): the Service Layer and the Topological Layer.

Deliverable 1.2: Time-Space Diagram

Input File: `stop_times.txt`

Task: For **one selected stop**, enumerate all “service nodes” (*stop, time*) between 7:00–9:00 AM

Output:

- Time-space diagram (hand-drawn or generated)
- Table listing: `trip_id`, `arrival_time`, `departure_time`, `route_id`
- Headway calculation between consecutive arrivals

Metric 1.2: Headway Deviation (Coefficient of Variation)

Let h_i be the headway (time gap) before arrival i , and $\bar{h}$ the mean headway:

$$CV_h = \frac{\sigma_h}{\bar{h}} = \frac{\sqrt{\frac{1}{n} \sum_{i=1}^n (h_i - \bar{h})^2}}{\bar{h}}$$

Interpretation:

CV Value	Service Quality
< 0.2	Well-regulated service
0.2–0.5	Moderate bunching
> 0.5	Severe bunching (operational problem)

Connection to Practice: In Lecture 2.1, Joe Gregory discusses how On-Time Performance suffers when drivers don't have enough run time or layover time. Your headway deviation metric quantifies this problem.

3 Phase 2: Operations & Optimization

This phase bridges data understanding to service planning decisions, directly applying concepts from Joe Gregory's Valley Metro presentation.

Required Reading for Phase 2

Primary: Lecture 2.1 — Valley Metro Service Planning (Joe Gregory)

Focus Areas:

- Slides 13–16: Scheduling/Blocking and Fiesta BUZZ Case Study
- Slides 17–21: Connecting to New Service (AV Pilot)

Secondary: Lecture 2.2 — Bus and Driver Scheduling

3.1 Module 2.1: Interlining & Vehicle Blocking

Concept: The Fiesta BUZZ Case Study (from Lecture 2.1)

From Lecture 2.1 (Slide 14):

- New route introduced in October 2022
- 52 minutes of runtime round trip
- 2 vehicles being used
- **Problem:** On-Time Performance suffers from insufficient run time or layover time

Solution (Slide 16):

- Interline routes (DBUZ and FBUZ)
- Add one vehicle
- Adjust dwell time locations

This case study demonstrates that `stops.txt` locations and `stop_times.txt` schedules are **decision variables**, not fixed constraints.

Deliverable 2.1: Deadhead Analysis

Task: Identify blocking opportunities in your GMNS network

Specific Requirements:

1. Find Trip A ending at time t_1 at stop s_1
2. Find Trip B starting at time $t_2 > t_1$ at stop s_2 nearby
3. Calculate:

- Deadhead distance: $d(s_1, s_2)$ in meters
- Layover time: $t_2 - t_1$ in minutes
- Is layover $\geq$ minimum required (typically 5–10 min)?

Optimization Question: If you relocate stop s_2 by 200m in `stops.txt`, does deadhead decrease? Can this save a vehicle?

3.2 Module 2.2: Frequency vs. Fleet Size Trade-off

The Fundamental Fleet Equation

From Lecture 2.2, the relationship between service frequency and fleet size:

$$N_{\text{buses}} = \frac{T_{\text{cycle}}}{h}$$

where:

$$T_{\text{cycle}} = T_{\text{running}} + T_{\text{layover}} \quad (\text{round-trip cycle time})$$

$$h = \text{headway (time between buses)}$$

$$N_{\text{buses}} = \text{number of vehicles required}$$

Implication: Doubling frequency ($h \rightarrow h/2$) doubles fleet requirement ($N \rightarrow 2N$).

Deliverable 2.2: Fleet Optimization Under Constraint

Given:

- A fixed fleet of $N = 10$ buses
- Current route with $T_{\text{running}} = 45$ min (one direction)
- Required layover: $T_{\text{layover}} = 8$ min at each terminal

Tasks:

1. Calculate cycle time: $T_{\text{cycle}} = 2 \times 45 + 2 \times 8 = 106$ min
2. Determine minimum achievable headway: $h_{\text{min}} = T_{\text{cycle}}/N = 10.6$ min
3. If you want 10-minute headway, how many buses do you need?
4. **Optimization:** Propose a schedule modification in `stop_times.txt` that achieves the highest frequency possible without exceeding 10 buses

Constraint: Your solution must maintain minimum layover times for driver breaks.

4 Phase 3: Demand, Load, and Congestion

This is where students prove the network is “real” by adding passengers. This phase directly implements the applications described in Lecture 2.4, Section 6.

Required Reading for Phase 3**Primary:** Lecture 2.4 — GTFS2GMNS User Guide, Section 6 (Applications)**Focus Areas:**

- Section 6.1: Traffic Assignment
- Section 6.5: Linear Programming Applications

Reference: Lecture 2.1, Slides 8–12: Origin and Destination Study**4.1 Module 3.1: Building the OD Layer**

From Lecture 2.1 (Joe Gregory), Valley Metro conducts O-D surveys every 3–5 years, collecting 17,000 completed surveys covering trip origin/destination, transfers, trip purpose, fare type, and access/egress modes.

Deliverable 3.1: Synthetic OD Matrix**Task:** Create a simple CSV representing 1,000 passenger trips**Required Fields:**

Field	Type	Description
trip_id	integer	Unique passenger trip identifier
origin_stop	string	Boarding stop (from <code>stops.txt</code>)
dest_stop	string	Alighting stop
dep_time	HH:MM	Desired departure time

Generation Method (choose one):

1. **Gravity Model:** $T_{ij} = k \cdot \frac{P_i \cdot A_j}{d_{ij}^2}$ where P_i = population near stop i , A_j = jobs near stop j
2. **Uniform Random:** Random selection of O-D pairs with peak-hour concentration
3. **Census-Based:** Use ACS commute data if available

4.2 Module 3.2: Load Profile and Congestion Detection

Deliverable 3.2: Load Profile Graph

Output: A graph showing passenger load along a route

Axes:

- X-axis: Distance along route (or stop sequence)
- Y-axis: Number of passengers on vehicle

Calculation: For each link (i, j) on a trip:

$$\text{Load}_{ij} = \text{Load}_{i-1,i} + \text{Boardings}_i - \text{Alightings}_i$$

Constraint Check:

$$\rho_{ij} = \frac{\text{Load}_{ij}}{\text{Capacity}} \leq 1.0$$

If $\rho > 1.0$ (overcrowding), student must propose solution.

Metric 3.1: Load Factor (ρ)

The load factor measures vehicle utilization:

$$\rho = \frac{\text{Passengers on arc}}{\text{Vehicle capacity}}$$

Interpretation and Action:

ρ Value	Status	Required Action
< 0.5	Underutilized	Consider reducing frequency
0.5–0.85	Healthy	Optimal operating range
0.85–1.0	Near capacity	Monitor closely
> 1.0	Overcrowded	Must solve: increase frequency or capacity

5 Phase 4: Measuring Network “Goodness”

To ensure students aren’t just doing “busy work,” they must compute and report standardized performance metrics.

Required Reading for Phase 4

Primary: Lecture 2.4 — Sections 6.3 (Accessibility) and 6.4 (Equity Analysis)

Reference: Lecture 2.1 — Slides 5–7: System Perspectives (Riders, Operators, Funders)

5.1 Three Essential Performance Statistics

Statistic 1: Stop Accessibility

Definition: Percentage of population within 400m walking distance of a transit stop.

$$\text{Accessibility} = \frac{\sum_{i \in \text{Stops}} \text{Pop}(\text{Buffer}_{400m}(i))}{\text{Total Population}} \times 100\%$$

Calculation Method:

1. Extract stop coordinates from `node.csv`
2. Create 400m buffer around each stop
3. Overlay with census population data
4. Compute coverage percentage

Benchmark: Good urban transit systems achieve > 80% coverage.

Statistic 2: Effective Speed

Definition: Average speed including all delays (dwell time, traffic, etc.)

$$\text{Effective Speed} = \frac{\text{Total Route Distance}}{\text{Total Travel Time (including dwell)}}$$

Threshold Rule:

- If Effective Speed < 10 mph, student must propose **Stop Consolidation**
- Stop Consolidation = deleting rows in `stops.txt` to reduce dwell time
- Must justify which stops to remove (low ridership, close spacing)

Statistic 3: Transfer Penalty

Definition: Average waiting time at transfer nodes (where multiple routes intersect)

$$\text{Transfer Penalty} = \frac{1}{|T|} \sum_{t \in T} (\text{Departure}_{\text{connecting}} - \text{Arrival}_{\text{arriving}})$$

where T is the set of all transfer opportunities.

Connection to GMNS: Transfer links in `link.csv` (`link_type` = “transferring”) connect physical stations. The penalty is encoded in `VDF_penalty1`.

Benchmark: Well-coordinated systems achieve transfer penalties < 10 minutes.

6 Deliverable Checklist and Grading Rubric

6.1 Summary Checklist (Template)

Step	Deliverable	What It Proves	Points
1	Clean GMNS CSVs	Data pipeline competency	15
2	Time-Space Diagram	Time-expanded logic understanding	15
3	Vehicle Requirement Calc	Schedule-Cost link	20
4	Congestion Heatmap	Network-Demand link	25
5	Optimization Memo	Problem-solving ability	25
Total			100

6.2 Detailed Rubric

Deliverable 1: Clean GMNS CSVs (15 points)

- (5 pts) `node.csv` contains correct physical and service nodes
- (5 pts) `link.csv` contains all four link types
- (5 pts) Data passes validation (no missing fields, valid geometries)

Deliverable 2: Time-Space Diagram (15 points)

- (5 pts) Correct identification of service nodes for selected stop
- (5 pts) Accurate headway calculations
- (5 pts) CV calculation and interpretation

Deliverable 3: Vehicle Requirement (20 points)

- (5 pts) Correct cycle time calculation
- (5 pts) Correct fleet size calculation
- (5 pts) Deadhead analysis with specific trip pairs
- (5 pts) Optimization proposal with quantified savings

Deliverable 4: Congestion Heatmap (25 points)

- (5 pts) Valid OD matrix with 1,000 trips
- (10 pts) Correct load profile calculation
- (5 pts) Identification of overcrowded segments
- (5 pts) Visualization quality

Deliverable 5: Optimization Memo (25 points)

- (5 pts) Clear problem statement
- (10 pts) Quantified before/after comparison
- (5 pts) Feasibility analysis (constraints respected)
- (5 pts) Professional writing quality

7 Connection to Advanced Topics

7.1 From Transit to Multi-Modal Systems

The framework presented here extends naturally to other transportation systems. The state-action-transition structure applies universally:

System	State s_t	Action a_t	Capacity Evolution
Transit	Platform queue, vehicle load	Dispatch, hold	Boarding constraints
Airport Ground	Gate occupancy, taxiway load	Pushback, ground hold	Runway capacity
Airspace	Sector occupancy	Metering, reroute	Sector capacity
Power Grid	Line loading, reserve	Generation dispatch	Thermal limits

7.2 Linear Programming Formulation

From Lecture 2.4 (Section 6.5), the GMNS network directly maps to LP models:

Transit Assignment Problem:

$$\min \sum_{a \in A} \int_0^{x_a} t_a(\omega) d\omega \quad (1)$$

$$\text{s.t.} \quad \sum_{p \in P_{rs}} f_p = q_{rs} \quad \forall (r, s) \in OD \quad (2)$$

$$x_a = \sum_{p \in P} \delta_{ap} f_p \quad \forall a \in A \quad (3)$$

$$x_a \leq u_a \quad \forall a \in A \quad (4)$$

$$f_p \geq 0 \quad \forall p \in P \quad (5)$$

where:

- A : Set of links from GMNS network
- x_a : Flow on link a
- t_a : Travel time (from `VDF_fftt1`)
- u_a : Capacity (from `capacity` field)

8 Appendix: Quick Reference

8.1 GTFS File Summary

File	Key Fields	GMNS Mapping
<code>stops.txt</code>	<code>stop_id</code> , <code>stop_lat</code> , <code>stop_lon</code>	Physical nodes
<code>routes.txt</code>	<code>route_id</code> , <code>route_type</code>	<code>directed_route_id</code>
<code>trips.txt</code>	<code>trip_id</code> , <code>route_id</code> , <code>direction_id</code>	Service links
<code>stop_times.txt</code>	<code>arrival_time</code> , <code>departure_time</code>	<code>VDF_fftt1</code>
<code>shapes.txt</code>	<code>shape_pt_lat/lon</code> , <code>shape_pt_sequence</code>	Link geometry

8.2 GMNS Link Types

Link Type	From Node	To Node	Purpose
Service	Service node	Service node	In-vehicle travel
Boarding	Physical	Service	Passenger boards
Deboarding	Service	Physical	Passenger alights
Transfer	Physical	Physical	Walk between stops

8.3 Key Formulas

Circuitry: $\frac{\text{Route Length}}{\text{Euclidean Distance}}$

Headway CV: $\frac{\sigma_h}{\bar{h}}$

Fleet Size: $N = \frac{T_{\text{cycle}}}{h}$

Load Factor: $\rho = \frac{\text{Passengers}}{\text{Capacity}}$

Effective Speed: $\frac{\text{Distance}}{\text{Total Time}}$

8.4 Reading Assignment Mapping

Phase	Primary Reading	Key Sections
Phase 1	Lecture 2.4 (GTFS2GMNS Guide)	Sections 1–4
Phase 2	Lecture 2.1 (Valley Metro)	Slides 13–21
Phase 2	Lecture 2.2 (Bus Scheduling)	All
Phase 3	Lecture 2.4 (GTFS2GMNS Guide)	Section 6
Phase 3	Lecture 2.1 (Valley Metro)	Slides 8–12
Phase 4	Lecture 2.4 (GTFS2GMNS Guide)	Sections 6.3–6.4